

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory

Alfacip 0.25 mcg

(Alfacalcidol capsules 0.25 mcg)

Composition

Each soft gelatin capsule contains

Alfacalcidol BP0.25 mcg

(1 α -Hydroxyvitamin D₃)

Approved colours used in capsule shell

Brilliant blue supra & Quinoline yellow

Description

Green transparent round shaped soft gelatin capsules.

Pharmacology

Pharmacodynamics

Alfacalcidol undergoes rapid hepatic conversion to 1,25-dihydroxyvitamin D₃ metabolite which acts as a regulator of calcium and phosphate metabolism which acts as a regulator of calcium and phosphate metabolism. The main effects are to increase circulating 1,25-dihydroxyvitamin D₃ levels and thereby to increase intestinal absorption of calcium and phosphate, promote bone mineralization, decrease plasma parathyroid hormone levels as well as to decrease bone resorption, with relief of bone and muscle pain. Treatment of intestinal calcium malabsorption associated with postmenopausal, senile and steroid induced osteoporosis, with alfacalcidol improves the negative calcium balance of these patients.

The more positive calcium balance, produced by alfacalcidol treatment in osteoporotic patients, decreases bone loss and bone fracture.

Pharmacokinetics

In patients with renal failure, 1-5 μ g/day of 1 α -hydroxyvitamin D (1 α -OHD₃) increased intestinal calcium and phosphorus absorption in a dose-related manner. This effect was observed within 3 days of starting the drug and conversely, it was reversed within 3 days of its discontinuation.

In patients with nutritional osteomalacia, increases in calcium absorption were

noted within 6 hours of giving 1 µg 1α-OHD3 orally and usually peaked at 24 hours. 1α-OHD3 also produced increases in plasma inorganic phosphorus due to increased intestinal absorption and renal tubular re-absorption. This latter effect is a result of PTH suppression by 1α-OHD3. The effect of the drug on calcium was about double its effect on phosphorus absorption.

Patients with chronic renal failure have shown increased serum calcium levels within 5 days of receiving 1α-OHD3 in a dose of 0.5 - 1.0 µg/day. As serum calcium rose, PTH levels and alkaline phosphatase decreased toward normal.

Indication

Alfacalcidol is used for treating conditions in which calcium metabolism is disturbed due to impaired 1-alpha hydroxylation, such as reduced renal function; in other disorders associated with Vitamin D resistance; and in calcium malabsorption of osteoporosis.

The main indications are:

1. Renal bone disease (renal osteodystrophy)
2. Hypoparathyroidism
3. Hypoparathyroidism (with bone disease)
4. Nutritional and malabsorptive rickets and osteomalacia
5. Hypophosphataemic Vitamin D-resistant rickets and osteomalacia
6. Pseudo-deficiency (D-dependent Type I) rickets with osteomalacia

Dosage & Administration

General: The dosage and administration of Alfacip should be adjusted as follows:

The initial dose is

All indications, excepting osteoporosis:

Adults	1mcg/day Elderly patients	0.5
mcg/day Children 20 kg & over		
Excepting in renal osteodystrophy	1mcg/day	
In renal osteodystrophy	0.04-0.08mcg/day	Children under
20 kg	0.05 mcg/kg/day	Osteoporosis
0.5 mcg/day		

Contraindications

Alfacalcidol should not be administered in the presence of hypercalcaemia, hyperphosphataemia (except when occurring with hypoparathyroidism) or hypermagnesaemia. Alfacalcidol should not be used in patients with evidence of Vitamin D toxicity or known hypersensitivity to the effects of Vitamin D or any of its analogues.

Warning and Precautions

Alfacalcidol increases the intestinal absorption of calcium and phosphate, serum levels of which should be monitored, particularly in patients with renal failure. Throughout treatment with Alfacalcidol regular plasma and urinary (24-hour collection) calcium levels should be determined at least once every three months. Alfacalcidol therapy requires regular monitoring of calcium phosphate, alkaline phosphate, magnesium and creatinine levels as well as other appropriate biochemical parameters and should only be prescribed when suitable facilities are available. If there is biochemical evidence of bone healing (eg. return towards normal serum alkaline phosphate levels), hypercalcaemia may develop if the dose of Alfacalcidol is not decreased appropriately. If hypercalcaemia or hypercalciuria occur, this can be corrected rapidly by stopping treatment with Alfacip and any calcium supplements until plasma calcium levels return to normal, usually in about a week.

Alfacalcidol should be administered with caution to patients with hypercalciuria especially those with a history of renal calculi.

Drug interactions

Alfacalcidol/Digitalis/Glycosides

Hypercalcaemia in patients taking digitalis preparations may precipitate cardiac arrhythmias. Patients taking digitalis concurrently with alfacalcidol must therefore be closely monitored.

Alfacalcidol/Barbiturates/Enzyme-inducing Anticonvulsant Drugs

Patients on barbiturates or other enzyme-inducing anticonvulsants may require an increased dose of alfacalcidol to produce the desired effect.

Alfacalcidol/Drugs affecting intestinal absorption

Absorption of alfacalcidol may be impaired by concurrent use of mineral oil (prolonged use), cholestyramine, colestipol, sucralfate or large amounts of aluminium-based antacids.

Alfacalcidol/Magnesium

Caution should be exercised in the use of magnesium-based antacids or laxatives for patients taking alfacalcidol who are on chronic renal dialysis. Hypermagnesaemia may occur.

Alfacalcidol/Calcium/Thiazides

The risk of hypercalcaemia is increased in patients taking calcium-containing preparations or thiazide diuretics concurrently with alfacalcidol.

Alfacalcidol/Vitamin D and Derivatives

Alfacalcidol is a potent derivative of Vitamin D. Pharmacological doses of vitamin D and its derivatives should not be given during alfacalcidol treatment because of the possibility of additive effects and an increased risk of hypercalcaemia.

Pregnancy and Lactation

Pregnancy

There is insufficient evidence on which to assess the safety of alfacalcidol use during pregnancy. Animal study has not revealed any hazard but as with all drugs Alfapip should only be used during pregnancy if treatment is essential and no better alternative is available.

Lactation

Although not definitely established, it is likely that increased levels of 1,25-dihydroxyvitamin D₃ will be found in the breast milk of mothers treated with alfacalcidol. This might have some influence on calcium metabolism in a breast-feeding infant and discontinuation of breast-feeding should be considered.

Side Effects

Adverse effects generally relate to hypercalcaemia and in the case of renal impairment, hyperphosphataemia which may be induced by alfacalcidol therapy.

Hypercalcaemia may manifest as malaise, fatigue, weakness, dizziness, headache, nausea, dry mouth, constipation, diarrhea, heartburn, vomiting, abdominal pain or other gastrointestinal discomfort, muscle pain, bone pain, joint pain, pruritus or palpitations. In hypercalcaemic dialysis patients, the possibility of calcium influx from the dialysate should be considered. No other side effects associated directly with alfacalcidol therapy have been noted

Overdosage

Overdosage of alfacalcidol can lead to hypercalcaemia which may manifest clinically as malaise, fatigue, weakness, dizziness, drowsiness, headache, nausea, dry mouth, constipation, diarrhea, heartburn, vomiting, abdominal pain or other gastrointestinal discomfort, muscle pain, bone pain, joint pain, pruritis or palpitations. Administration of alfacalcidol should be stopped if hypercalcaemia occurs. Severe hypercalcaemia may require treatment with general supportive measures, with intravenous fluids, and if needed, with a loop diuretic or corticosteroids.

In acute overdosage, early treatment with gastric lavage and/or the administration of mineral oil may reduce absorption and promote faecal elimination.

Storage

Store below 30° C

Shelf life

2 years

Packing/Pack Size

Carton containing 10 blisters of 10 capsules each

Registration holder in Malaysia

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